

Assignment 7

October 4, 2026

Question 1

(a)

we are given:

- intensity of the light: $I = 1 \text{ W m}^{-2}$
- binding energy: $E_{\text{bind}} = 1.8 \text{ eV}$
- Bohr radius: $r = 5.3 \times 10^{-11} \text{ m}$

we assume that the atom absorbs all the radiation that falls within its Bohr radius.

the area of the atom is approximately the area of a circle,

$$A = \pi r^2$$

substituting,

$$A = \pi(5.3 \times 10^{-11})^2$$

$$A \approx 8.82 \times 10^{-21} \text{ m}^2$$

the intensity is power per unit area,

$$I = \frac{P}{A}$$

so,

$$P = IA$$

substituting,

$$P = (1)(8.82 \times 10^{-21})$$

$$P = 8.82 \times 10^{-21} \text{ W}$$

the binding energy is given in eV, so we convert it to joules.

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

therefore,

$$E_{\text{bind}} = 1.8(1.602 \times 10^{-19})$$

$$E_{\text{bind}} \approx 2.884 \times 10^{-19} \text{ J}$$

using

$$E = Pt$$

$$t = \frac{E}{P}$$

substituting,

$$t = \frac{2.884 \times 10^{-19}}{8.82 \times 10^{-21}}$$

$$t \approx 32.7 \text{ s}$$

hence,

$$\boxed{t \approx 33 \text{ s}}$$

so, it would take about 33 seconds to collect enough energy to eject an electron.

(b)

the calculation in part (a) treats light as a continuous wave, so the electron slowly builds up energy until it has enough to escape.

in reality, light comes in packets called photons, each with energy $E = hf$. an electron absorbs one whole photon at a time. if one photon has enough energy ($hf \geq \phi$), the electron is ejected immediately.

making the light weaker only means fewer photons arrive, not that each photon has less energy. so emission still starts instantly.

Question 2

(a)

we have light travelling from a medium with refractive index

$$n_1(\lambda)$$

into a medium with

$$n_2 = 1.$$

using Snell's law,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

since $n_2 = 1$,

$$n_1 \sin \theta_1 = \sin \theta_2$$

therefore,

$$\theta_2 = \arcsin(n_1 \sin \theta_1)$$

we want to find how much the refracted angle changes when the refractive index changes.

differentiate with respect to n_1 ,

$$\boxed{\frac{d\theta_2}{dn_1} = \frac{\sin \theta_1}{\sqrt{1 - n_1^2 \sin^2 \theta_1}}}$$

as θ_1 increases, the denominator becomes smaller, so the change in θ_2 becomes larger.

however, the light must still be transmitted into the second medium.

from Snell's law,

$$\sin \theta_2 = n_1 \sin \theta_1$$

and since $\sin \theta_2 \leq 1$, we have

$$n_1 \sin \theta_1 \leq 1$$

the maximum possible value occurs when

$$n_1 \sin \theta_1 = 1$$

$$\sin \theta_1 = \frac{1}{n_1}$$

$$\theta_1 = \arcsin\left(\frac{1}{n_1}\right)$$

this is the critical angle.

hence $\boxed{\theta_1 = \theta_c}$ is the best choice of incident angle to maximise the dispersion.

at this angle, $\theta_2 = 90^\circ$

(b)

we are given:

$$n_{400} = 1.8$$

and

$$n_{700} = 1.7$$

the incident angle should be the critical angle for the larger refractive index, since both wavelengths need to be transmitted.

from part (a),

$$\theta_1 = \arcsin\left(\frac{1}{1.8}\right)$$
$$\theta_1 \approx 33.75^\circ$$

- for the 400 nm light,

$$1.8 \sin(33.75^\circ) = \sin \theta_{2,400}$$

since 33.75° is the critical angle,

$$1.8 \sin(33.75^\circ) = 1$$

so,

$$\theta_{2,400} = 90^\circ$$

- for the 700 nm light,

$$1.7 \sin(33.75^\circ) = \sin \theta_{2,700}$$

$$\sin \theta_{2,700} = 1.7(0.5556)$$

$$\sin \theta_{2,700} \approx 0.9444$$

$$\theta_{2,700} = \arcsin(0.9444)$$

$$\theta_{2,700} \approx 70.8^\circ$$

the angle between the two ends of the optical spectrum is therefore

$$\Delta\theta = 90^\circ - 70.8^\circ$$

$$\Delta\theta \approx 19.2^\circ$$

$$\boxed{\Delta\theta \approx 19^\circ}$$

Question 3

the thin lens equation is

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

since the sun is very far away,

$$u \rightarrow \infty$$
$$\frac{1}{u} \rightarrow 0$$

so,

$$\frac{1}{f} = \frac{1}{v}$$

$$\boxed{v = f}$$

so the image of the sun forms one focal length behind the lens.
the sun has an angular diameter of

$$0.5^\circ$$

so the angle from the centre of the sun to its edge is

$$\theta = \frac{0.5^\circ}{2} = 0.25^\circ$$

consider the ray from the centre of the lens to the edge of the solar image.
using the right triangle,

$$\tan \theta = \frac{r}{f}$$
$$r = f \tan \theta$$

substituting $\theta = 0.25^\circ$,

$$r = f \tan(0.25^\circ)$$

the diameter is

$$D = 2r$$

therefore,

$$D = 2f \tan(0.25^\circ)$$

$$D = 0.008726f$$

$$\boxed{D \approx 8.73 \times 10^{-3} f}$$

where D and f are in the same units.